Seyed Soheil Johari

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Professional Summary

Ph.D. candidate in Computer Science at the **University of Waterloo** with one year of full-time experience as a **Data Center Networking Researcher** at **Huawei Technologies**. My expertise spans networking systems design and performance optimization for **large-scale AI infrastructure**, **distributed training** environments, and LLM-driven workloads, along with **ML-driven automation** and **fault management** in **cloud-native**, softwarized **5G network** slice architectures.

Education

Ph.D. in Computer Science, ***University of Waterloo*** **2021 — 2026**

B.Sc. in Electrical Engineering, ***Sharif University of Technology*** **2015 — 2020**

Technical and Industrial Experience

DCN Researcher: AI Infrastructure Networking
Huawei Technologies **May 2025 — April 2026**

- Optimization of **large-scale AI training GPU cluster** networking for **Large Language Model (LLM)** training workloads, focusing on load balancing, **transport-layer** mechanisms, and host/switch-level control in **RDMA-based** data center environments.
- Designed a new **load balancing** mechanism for ACK coalescing scenarios in AI training networks, improving **bandwidth utilization by up to 40%** under **link failure** scenarios compared to **UEC-standardized** schemes. **patent application under review.**
- Designed an **in-switch load balancing architecture** at the **destination core switch**, separating source-DC and destination-DC decisions and improving **bandwidth utilization by up to 24%** under **link failure** scenarios. **patent application under review.**
- Designed a novel **credit-based host-side congestion control** mechanism for AI training clusters, improving **flow completion time by up to 15%** under bursty **distributed training** workloads.
- Designed realistic **AI training and inference workloads** for Huawei's AI training environment to evaluate transport-layer mechanisms (**all-reduce**, **bursty gradients**, heterogeneous flows, pipeline **parallelism**, and **latency-sensitive inference**).
- Contributed to the design and refinement of *HT-CC*, a lightweight **congestion control** algorithm optimized for ease of deployment in **Huawei AI clusters**, focusing on simplicity, stability, and practical implementability.

Research Assistant: AI-driven 5G Network Slice Operations and Management
University of Waterloo **January 2021 — April 2025**

- Project page: https://rboutaba-cs.github.io/wat5gslicing/
- Conducted research in collaboration with Rogers Communications on **AI-driven 5G network automation**, focusing on automated **fault detection, analysis, and resolution** in **5G network slices** built on **cloud-native** and **NFV**-based infrastructures.
- Proposed a **causal RCA framework** for 5G and cloud-native systems, learning a **latent-aware structural prior** from normal-operation telemetry to guide root cause discovery under limited data; improving RCA accuracy by up to **38%**.
- Designed a **few-shot domain adaptation** framework for 5G network management, combining **causal** feature separation and **GAN**-based reconstruction to mitigate data drift; achieving up to **44% improvement** over state-of-the-art methods.
- Developed an **Active Learning** framework for **Transformer**-based fault diagnosis in 5G networks, selecting informative samples via attention dependency graphs; maintaining accuracy with **50% fewer labels** and up to **150% better unseen-fault detection**.
- Proposed a robust **anomaly detection and localization** framework for NFV systems using **masked autoencoders**, **Noisy-Student training**, and **XAI**; improving detection by up to 24% and localization accuracy by 22% across multiple NFV systems.
- Developed **Deep Reinforcement Learning**-based re-optimization mechanisms for network slice embedding in **Elastic Optical Networks** to reduce spectrum fragmentation while minimizing reconfiguration overhead.

MITACS Research Intern: ML-driven management of 5G Networks
Rogers Communications **January 2021 — April 2024**

- Contributed to **industry-facing** implementation and validation of AI-driven **fault and performance management** solutions for softwarized and virtualized 5G network slices in collaboration with Rogers engineers.
- Processed and analyzed real and emulated **5G core network telemetry** (KPIs, logs, alarms, resource metrics) to support model development, evaluation, and benchmarking under **realistic operational** scenarios.
- Implemented end-to-end **data pipelines** and experimental frameworks for anomaly detection, fault diagnosis, and root cause analysis in cloud-native NFV-based environments.

Skills

Networking Data Center Networking (AI/ML Clusters), RDMA/RoCEv2, InfiniBand, NVLink, Transport Protocol Design, Ultra Ethernet (UEC concepts), NS3, Kubernetes, Docker, NCCL, SDN, NFV, Network Softwarization, Linux Networking, ONOS, OpenStack, programmable switches, Clos / Fat-tree topologies,

Machine Learning Distributed Training Optimization for LLM Workloads and Large-Scale AI Clusters, Communication-Efficient ML (AllReduce, Collective Optimization), Applied Machine Learning for Network Management, Anomaly Detection, Root Cause Analysis, Data-Driven Fault Diagnosis, Reinforcement Learning, Active Learning, Causal Inference, Domain Adaptation, Adversarial Learning, Generative AI, Graph Analysis, Geometric Deep Learning

Programming C, C++, Python, PyTorch, PyTorch Geometric, TensorFlow, Pandas, Java, Git, MATLAB, VHDL, Verilog

Publications

- **S. S. Johari**, R. Boutaba. "Root Cause Analysis in Networked Systems via Latent Cause Inference and Prior Knowledge." *Submitted to ACM SIGMETRICS*. Under Review.
- **S. S. Johari**, M. Tornatore, R. Boutaba, A. Saleh. "Few-Shot Domain Adaptation for Effective Data Drift Mitigation in Network Management." *IEEE International Conference on Distributed Computing Systems (ICDCS) 2025.* [pdf]
- **S. S. Johari**, N. Shahriar, M. Tornatore, R. Boutaba, A. Saleh. "Anomaly Detection and Localization in NFV Systems by Utilizing Masked-Autoencoder and XAI." *IEEE Transactions on Mobile Computing (TMC) 2025.* [pdf]
- **S. S. Johari**, N. Shahriar, M. Tornatore, R. Boutaba, A. Saleh. "Active Learning for Transformer-Based Fault Diagnosis in 5G and Beyond Mobile Networks." *IEEE Transactions on Network and Service Management (TNSM) 2025.* [pdf]
- **S. S. Johari**, S. Taeb, N. Shahriar, S. R. Chowdhury, M. Tornatore, R. Boutaba. "DRL-Assisted Reoptimization of Network Slice Embedding on EON-enabled Transport Networks." *IEEE Transactions on Network and Service Management (TNSM) 2023.* [pdf]
- **S. S. Johari**, N. Shahriar, M. Tornatore, R. Boutaba, A. Saleh. "Anomaly Detection and Localization in NFV Systems: An Unsupervised Learning Approach." *IEEE/IFIP Network Operations and Management Symposium (NOMS) 2022.* **Recipient of the Best Student Paper Award.** [pdf]